At N=24, narrowing the MLP 4x costs nothing at all.

Not a layer on the gate, and not one instance in 1500.

L* is the smallest num_layers reaching 0.90 on depth-T instances at fixed realized density. At N=24 the wide knee is 3 (finding 11). This job trained the thin cells that decide the knee at ff=256: L=3 clears 4 of 4 rho=0.60 bins, so L*(24, ff=256) = 3 and the width factor at N=24 is 1.00x.

THE GATE AND THE POOL AGREE, which they did not have to. On 1500 paired instances the thin L=3 scores 0.9487 against the wide L=3's 0.9500 -- -0.0013, 50 won / 52 lost, p=0.92. A coarse gate passing a cell is weak evidence; a paired test failing to separate it from the reference on 1500 instances is the strong form.

THE GRID IS NOW COMPLETE: six cells, three problem sizes, two MLP widths. The width cost on network depth is 1.00x at N=24, 1.50x at N=48 and above 1.50x at N=96 -- zero at the small end and growing. hshunt's question (does the reasoning depth a problem needs relate to the network depth it needs?) now has an architecture axis on it, and the answer is that the two knees diverge as the problem gets bigger.

job 50007138, COMPLETED 00:34:11, ExitCode 0:0 | scripts/slurm/sudoku2_n24_ff256_lstar.sbatch | commit e65b97e (L=3 cell) and 801c593 (L=2 cell + evals; warmup code, a no-op at warmup_epochs=0) | seed 0 | shuffle_adj=False, so every number here is a DECODING score

What the model actually sees

Every token is one categorical id into a 34-symbol vocabulary (N=24, q=4, degree=5). No text, no digits-as-characters.

VOCABULARY

0-5	BEGIN_HINTS BEGIN_GRAPH BEGIN_REASONING BEGIN_ROUND SOLVED UNRESOLVED	*section + round markers*
6-9	COLOR_0 COLOR_1 COLOR_2 COLOR_3	*the q colours*
10-33	N0 N1 ... N23	*one id per node, no structure in the id*

GRAMMAR

BEGIN_HINTS	(NODE COLOR)*
BEGIN_GRAPH	(NODE (NODE){degree})*
BEGIN_REASONING	(BEGIN_ROUND (NODE COLOR)+)*
SOLVED	

The graph section is a neighbour table, not an image: node, then its degree neighbours. THE ORDER WITHIN A ROW IS THE LEAK of finding 16.

THIS INSTANCE, 204 TOKENS (seed 50000000038)

HINTS	BEGIN_HINTS N0 COLOR_2 N1 COLOR_1 N4 COLOR_1 N5 COLOR_1 N7 COLOR_2 N9 COLOR_0 ... (23 tokens)
GRAPH	BEGIN_GRAPH N0 N17 N9 N6 N13 N21 N1 N6 N12 N7 N19 N14 ... (145 tokens)
REASONING	BEGIN_REASONING BEGIN_ROUND N3 COLOR_3 N21 COLOR_3 N23 COLOR_3 BEGIN_ROUND N6 COLOR_0 N15 COLOR_1 ... (36 tokens)

Loss is on the REASONING section only: 8 rounds, 13 deductions, 26 target tokens. Everything before BEGIN_REASONING is the problem statement.

spin_gnn.sudoku_tokens.Vocab + encode_instance; shuffle_adj=False

One real reasoning trace, at N=24

11 hints of 24 nodes, then 8 rounds of simultaneous deduction. Hints: N0=2 N1=1 N4=1 N5=1 N7=2 N9=0 N10=2 N12=0 ...

THE TRACE, AS EMITTED

```
BEGIN_ROUND          N3 COLOR_3 N21 COLOR_3 N23 COLOR_3
  3 deduced
BEGIN_ROUND          N6 COLOR_0 N15 COLOR_1
  2 deduced
BEGIN_ROUND          N11 COLOR_3 N13 COLOR_0
  2 deduced
BEGIN_ROUND          N18 COLOR_2 N20 COLOR_0
  2 deduced
... 4 further rounds

SOLVED
```

WHY ONE DEDUCTION FIRES

node N3, neighbours [17, 5, 4, 10, 19]

colours known before this round:

```
   [0, 1, 2]  (3 = q-1)
```

only colour left: 3

```
so the trace emits  N3 COLOR_3
```

The model is never shown this reasoning -- only the node/colour pair.

A round deduces every node whose neighbours already cover q-1 colours, all at once. Reasoning depth T is the number of rounds: a deduction in round 3 depends on ones made in rounds 1 and 2, which is the serial computation that network depth has to supply.

shuffle_adj=False here, so each node's neighbour row is still sorted by the neighbour's TRUE colour -- finding 16's leak. That is deliberate: this run extends the leaked-order L* ladder rather than starting a second one.

T=8 rounds, the 90th percentile of 40 draws from this config (range 1-10); seed 50000000038, shuffle_adj=False, shuffle_events=False; evidence from sudoku_tokens.evidence_rounds (pre-round state, as DD saw it)

The completed grid: L* vs (N, ff)

dim_feedforward is the hidden width of the per-layer MLP: at d_model=256, ff=1024 is the usual 4x expansion and ff=256 is 1x. Nothing else differs within a row.

	L* at ff=1024	L* at ff=256	width factor	source
N=24	3	3	1.00x	THIS RUN (50007138)
N=48	4	6	1.50x	finding 14 (49509466/49587538)
N=96	8	> 12	> 1.50x	finding 25 (49816217)

cell	n_params	val_nll	pool rate	bins >= 0.90
L=2, ff=1024	1,651,234	1.1407	not pooled	0 of 4
L=3, ff=1024	2,440,994	1.0173	0.9500	4 of 4
L=2, ff=256 (new)	863,266	1.1680	0.3793	0 of 4
L=3, ff=256 (new)	1,259,042	1.0150	0.9487	4 of 4
L=4, ff=256	1,654,818	0.9744	0.9587	4 of 4
L=6, ff=256	2,446,370	0.9470	0.9767	4 of 4

TWO MATCHED-PARAMETER PAIRS

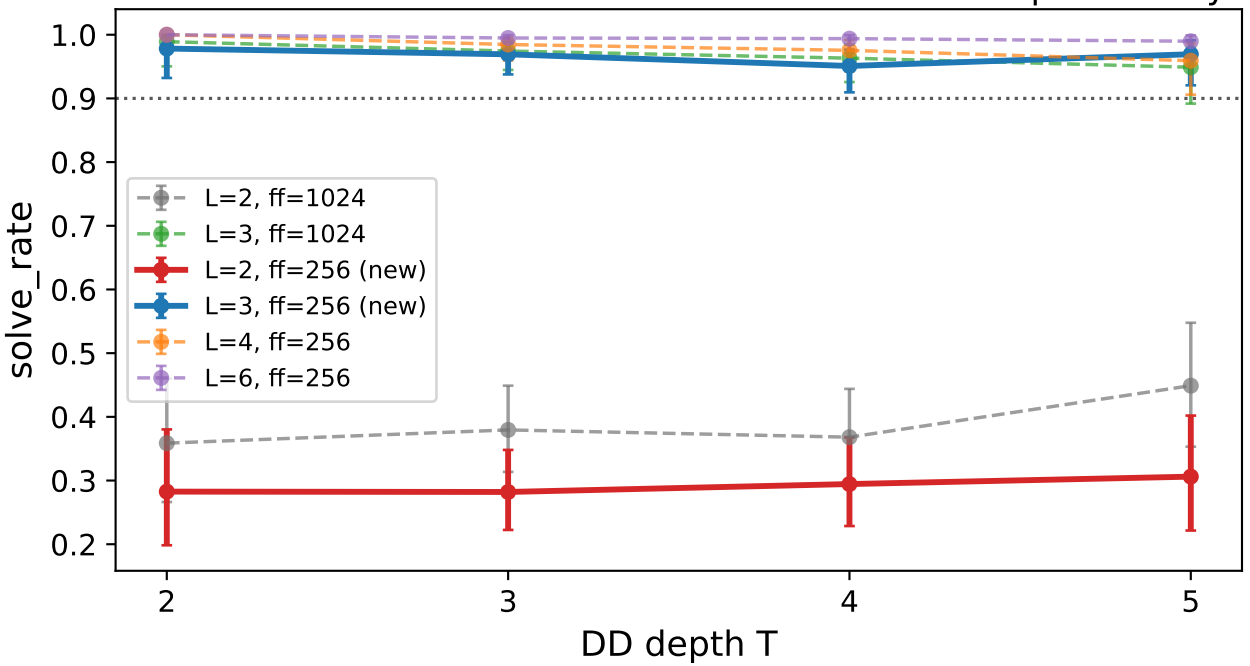
~1.65M, 0.2% apart: the wide shallow L=2 pools 0.385 over the four bins, the thin deep L=4 pools 0.980 -- +0.595, Fisher p=1e-116.
~2.44M, 0.2% apart: the thin deep L=6 beats the wide L=3 by +0.0267 on the n=1500 pool (65 won / 25 lost, p=3.0e-05), replicating finding 13 on an instance pool it was not measured on.

And the thin L=3 ties the wide L=3 (-0.0013, p=0.92) at 0.52x the parameters. At N=24 the MLP is not where this task's capacity goes.

L=4 and L=6 at ff=256 are job 49510197's checkpoints, re-used here at no cost

Four depth bins at rho=0.60, identical row for row

Realized density is identical across bins within a column, so what varies along the x-axis is the number of DD rounds the instance needs and nothing else. Every column was scored with --rho 0.75 0.60 --n-pool 600 --min-per-bin 40, so the bins come out row for row identical and the columns compare bin by bin.



	T=2	T=3	T=4	T=5	cleared	CA p
L=2, ff=1024	0.359	0.379	0.368	0.449	0/4	0.270
L=3, ff=1024	0.989	0.974	0.963	0.949	4/4	0.088
L=2, ff=256	0.283	0.282	0.294	0.306	0/4	0.661
L=3, ff=256	0.978	0.969	0.951	0.969	4/4	0.510
L=4, ff=256	1.000	0.985	0.975	0.959	4/4	0.037
L=6, ff=256	1.000	0.995	0.994	0.990	4/4	0.359

n per bin: T=2:92, T=3:195, T=4:163, T=5:98

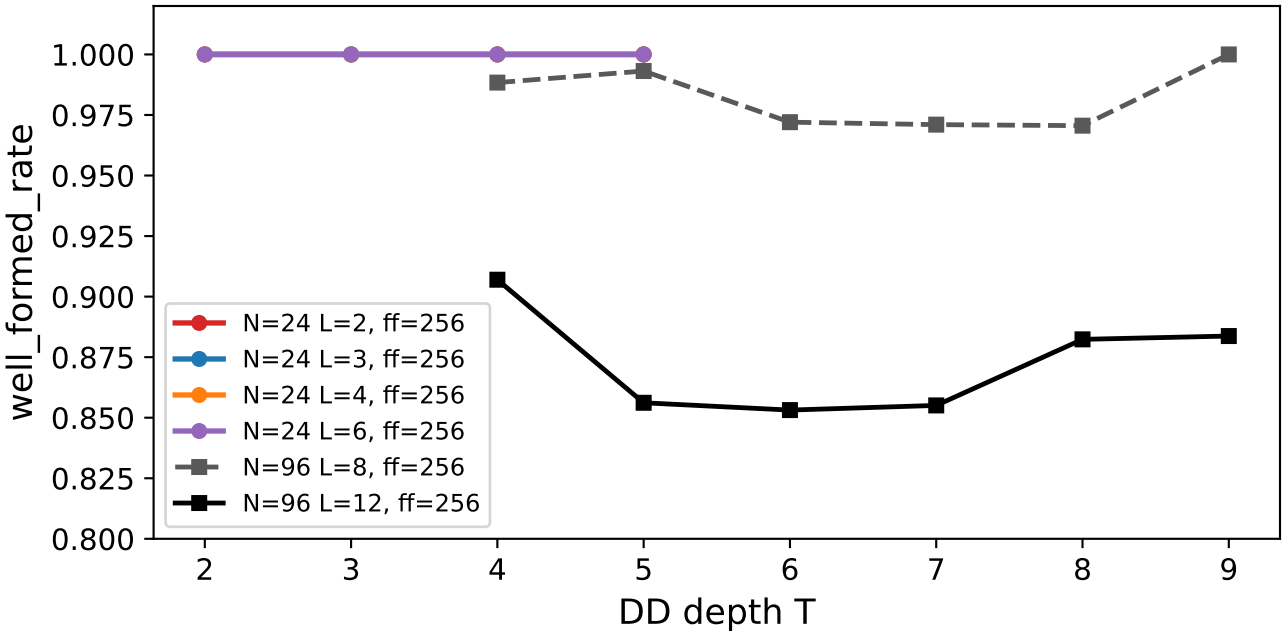
THE GATE IS A COARSE INSTRUMENT, DELIBERATELY

L* asks one question of each column -- does every bin reach 0.90 -- and it is the same question findings 11, 14, 22 and 25 asked, which is the only reason the six cells of the grid can be read against each other. It cannot distinguish 0.901 from 0.999, and at N=24 most bins sit near the ceiling: the Cochran-Armitage column is there to show how little room the trend statistic has here, not as a result.

beta 95% intervals per bin; Cochran-Armitage trend over the ordered bins, the statistic findings 11, 22 and 25 report

Does the thin MLP break the output grammar here too?

At N=96, L=12/ff=256 kept deducing (conditional solve rate 0.92-0.98, RISING with T) while only 0.853 of its generations parsed as a trace at all -- finding 25's unexpected mechanism. If a thin MLP simply breaks the grammar, it should break it at N=24 too. The plot is well_formed_rate for every thin cell at both sizes.



	worst wf	solve|wf 1st	solve|wf last	r vs T
N=24 L=2, ff=256	1.000	0.283	0.306	+0.94
N=24 L=3, ff=256	1.000	0.978	0.969	-0.50
N=24 L=4, ff=256	1.000	1.000	0.959	-0.99
N=24 L=6, ff=256	1.000	1.000	0.990	-0.97
N=96 L=8, ff=256	0.971	0.976	0.953	-0.38
N=96 L=12, ff=256	0.853	0.923	0.974	+0.72

PRE-REGISTRATION (d): CONFIRMED

The worst well_formed_rate anywhere in the N=24 thin column is 1.000, across all 16 bins, against 0.853 at N=96. The generative collapse is NOT a property of ff=256: it takes a deep thin stack at a large problem size, and at N=24 the thin cells write valid traces throughout. Whatever the width cost is here, it is a deduction cost, which is what makes the gate reading meaningful at this size.

well_formed_rate from sudoku_depth_eval; the N=96 comparison is finding 25 (job 49816217)

The metric of record: one pool, five checkpoints

1500 instances at seed_start=5e10, disjoint from every training set, reconstructed identically for all five cells (same N, same INSTANCE_KEYS), so the comparisons are paired instance for instance and McNemar's exact test applies.

cell	solve_rate	Wilson 95%	vs wide L=3	won/lost	McNemar p	outcome acc
L=3, ff=1024	0.9500	[0.9378, 0.9599]	reference	--	--	1.0000
L=2, ff=256 (new)	0.3793	[0.3551, 0.4042]	-0.5707	9/865	1.3e-242	1.0000
L=3, ff=256 (new)	0.9487	[0.9363, 0.9587]	-0.0013	50/52	9.2e-01	1.0000
L=4, ff=256	0.9587	[0.9474, 0.9676]	+0.0087	55/42	2.2e-01	1.0000
L=6, ff=256	0.9767	[0.9677, 0.9832]	+0.0267	65/25	3.0e-05	0.9993

PRE-REGISTRATION (b): FALSIFIED

The thin L=3 ties the wide L=3 on 1500 paired instances (-0.0013, p=0.92, 50 won / 52 lost). At N=24, MLP width buys literally nothing -- not a layer on the gate and not an instance on the pool. That is the clean zero the N=48 and N=96 width costs need to be read against.

PRE-REGISTRATION (c), THE LADDER FLOOR: CONFIRMED

L=2/ff=256 clears 0 of 4 bins and scores 0.290 pooled over the four rho=0.60 bins against the wide L=2's 0.385 (Fisher p=0.001, unpaired and therefore conservative). The claim held -- the thin L=2 does not beat the wide one -- but the REASON given for it did not: width is worth 9.5 points down here while being worth nothing at all at L=3. MLP width helps exactly where the stack is too shallow to deduce, and stops mattering once it is deep enough.

sudoku_agg_eval --n-instances 1500 --seed-start 50000000000, wide L=3 listed first

Pre-registrations, as resolved

Written into scripts/slurm/sudoku2_n24_ff256_lstar.sbatch before the job ran, with a named falsifier each.

	claim	verdict	evidence
(a)	L=3/ff=256 clears all 4 rho=0.60 bins, so L*(24, ff=256) = 3	CONFIRMED	clears 4 of 4; L*(24, ff=256) = 3, width factor 1.00x against the wide knee of 3.
(b)	On the n=1500 pool it is still resolvably WORSE than L=3/ff=1024	FALSIFIED	0.9487 vs 0.9500, -0.0013, 50/52 discordant, p=9.2e-01.
(c)	L=2/ff=256 clears no bin and does not beat the wide L=2	CONFIRMED	clears 0 of 4; pooled 0.290 vs 0.385, Fisher p=0.001.
(d)	well_formed_rate stays >= 0.95 in both new cells	CONFIRMED	worst well_formed_rate in the thin column is 1.000 (N=96's was 0.853).

3 OF 4 CONFIRMED.

The grid's width factor is the only one of the four that could have changed the shape of the story, and (b) is the one that decides whether 'no layers' means 'no cost'. The other two are controls: a ladder floor that is measured rather than assumed, and a check that finding 25's grammar collapse does not travel down to N=24 and invalidate the gate reading everywhere.

Each claim had a falsifier written beside it in the sbatch header; the header is the pre-registration of record

What this changes, and what comes next

CHANGED

- The L* vs (N, ff) grid is complete: 3/3 at N=24, 6/4 at N=48, >12/8 at N=96. The MLP-width cost on network depth is now a curve in N, not a single number from one size.
- Every bare 'L*(N)=k' in STATUS.md still means 'at ff=1024', and the grid now shows how much that caveat is worth at each size.
- The thin column at N=24 is the only complete thin ladder in the project (L=2, 3, 4, 6 all measured on the same bins), so it is the reference for what a thin ladder looks like when nothing is going wrong.

NOT CHANGED

- These are DECODING scores. shuffle_adj=False, so each node's adjacency row still arrives sorted by the neighbour's true colour (finding 16), and the whole L* ladder lives in the leaked task. Finding 23 established the budget a leak-free ladder needs (240000 instances x 100 epochs); it has not been run.
- One seed per cell. L* is read off a 0.90 gate on four bins at this size, and at N=24 most bins sit near the ceiling, so the gate has little room to resolve anything finer than 'clears' and 'does not'.
- The width factor across rows is not a paired comparison: each knee is measured on its own ladder at its own N, so the factor column carries no p-value.

NEXT, IN ORDER OF INFORMATION PER GPU-HOUR

1. Job 50008605 is already running: L=12 and L=8 at ff=256, N=96, with a 5-epoch lr warmup as the only change from finding 25. It decides whether 'L*(96, ff=256) > 12' is about computation or about trainability -- the one confound in the grid's most extreme cell.
2. L=10/ff=1024 at N=96 tightens the wide knee from 'between 6 and 8' to one rung.
3. The leak-free L*(T) ladder at N=48, at 240000 x 100ep -- the honest version of this whole grid, and the most expensive thing on the list.

checkpoints: d0256_h04_L03_ff00256_lr1e-03_pre and d0256_h04_L02_ff00256_lr1e-03_pre in sudoku_n24_ff256_2026-10-28